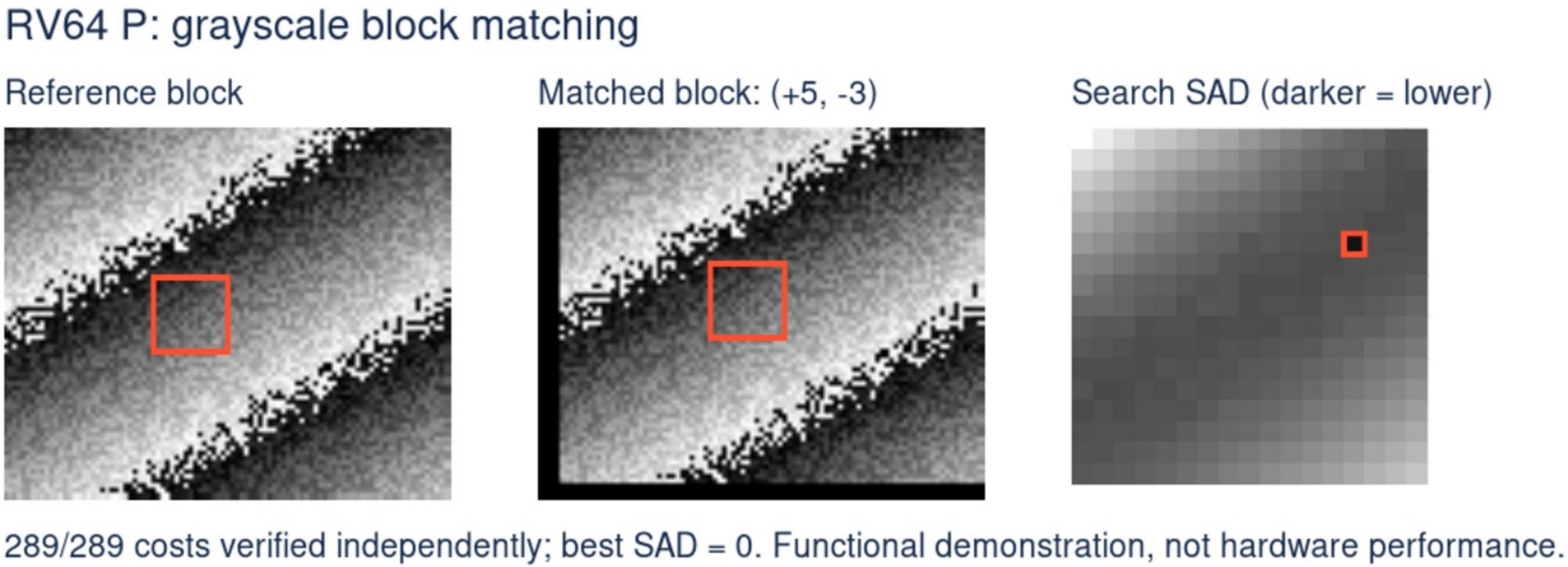
Image block matching with RV64 P

SAD = sum of absolute pixel differences. The lowest cost identifies the match.



Eight grayscale pixels per packed operation

The image kernel is C. No inline assembly is required.

```
#include <riscv_packed_simd.h>

for (; width - x >= 8; x += 8)
  {
    uint8x8_t av =
      __riscv_pld_u8x8 ((uint8_t *) (a + x));
    uint8x8_t bv =
      __riscv_pld_u8x8 ((uint8_t *) (b + x));
    sum += __riscv_pabdsumu_u8x8_u64 (av, bv);
  }
```

LOAD

Read eight pixels from each image.

REDUCE

Sum eight absolute byte differences.

ACCUMULATE

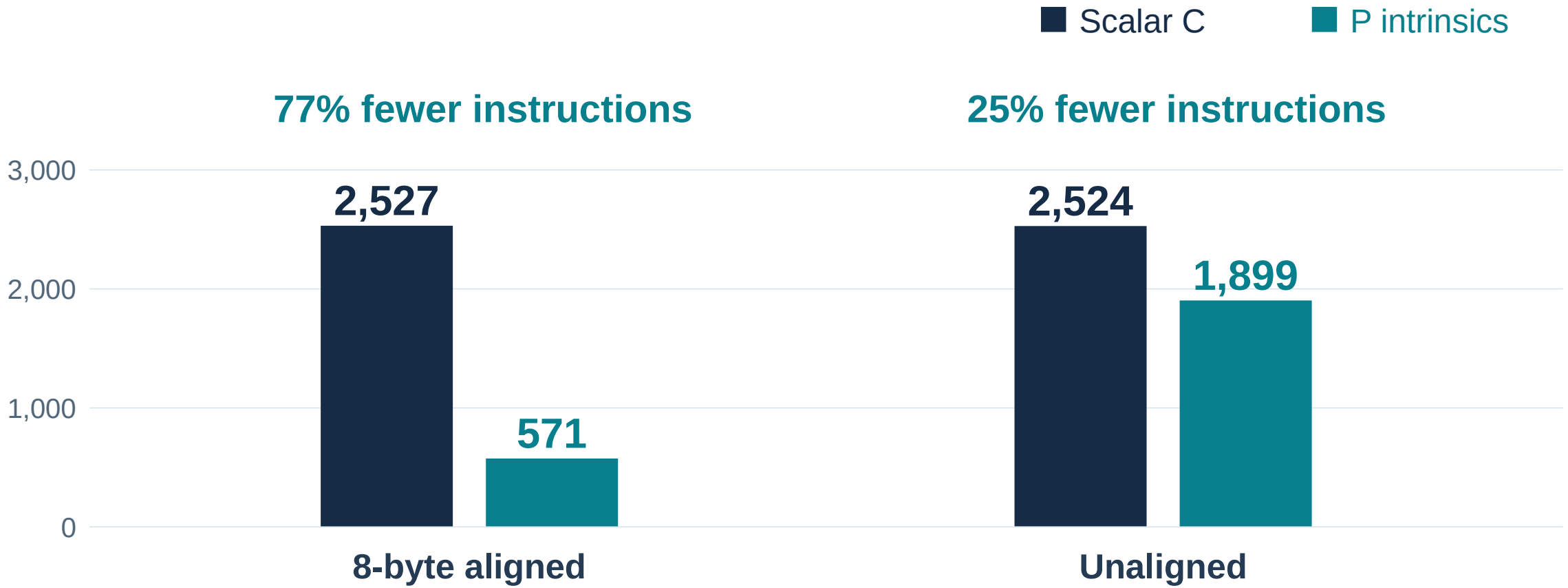
Add the result to a 64-bit total.

Generated instruction: pabdsumu.b

The full C kernel handles alignment, row strides and scalar tails.

Fewer instructions for image SAD

Executed QEMU guest instructions per 16 × 16 kernel call · -O3 · lower is better



Same compiler and common flags; no auto-vectorization or LTO.

Instruction counts are not cycles or hardware speedup.

From C source to a reproducible demo

P-enabled GCC, Binutils and QEMU support this static RV64 C example.

C source
Scalar + P intrinsics

→

GCC + Binutils
Compile and link

→

RV64 binary
Static Linux / lp64d

→

QEMU
Execute and trace

82,576
C checks per build

289 / 289
Image search costs matched

O2 · O3 · Os
All three builds pass

github.com/pz9115/RVP-Example

C code, build instructions and notebook

RUN AFTER BUILDING THE TOOLS

```
bash scripts/present.sh
```